

Homework_4

I first went to the ENCODE website to download RNA-seq data for the K562 cell line. I downloaded paired-end reads for the library **ENCLB822JYE**:

- **Reads1**: Accession ID [ENCFF002DKA](#)
- **Reads2**: Accession ID [ENCFF002DKE](#)

Next, I downloaded RNA-seq data for the MCF-7 cell line, specifically for the library **ENCLB555AQN**:

- **Reads1**: Accession ID [ENCFF000HQR](#)
- **Reads2**: Accession ID [ENCFF000HQP](#)

After downloading the RNA-seq files for both K562 and MCF-7, I installed **Cufflinks** using `wget` [Mac OS X](#) .

I also downloaded the file [hg19.ncbiRefSeq.gtf.gz](#), decompressed it, and renamed it to **hg19_ncbiRefSeq_annotation.gtf**. This annotation file is used for calculating gene expression and differential gene expression.

With all the necessary files ready, I began the alignment process for **K562 and MCF-7** using **Bowtie2**. After alignment, I used **Samtools** to filter uniquely mapped reads, sort, and index the BAM files. This step is necessary before running Cufflinks.

For this assignment, I used **MCF-7** as an example because I already had the K562 data from my second assignment. I am uploading those screenshots as well.

Screenshots of K562 & MCF-7 Alignment by using the Bowtie2

```
Mohana_Alla 🐼🐼🐼 Bio_App_new % cd ..
Mohana_Alla 🐼🐼🐼 Desktop % cd ..
Mohana_Alla 🐼🐼🐼 ~ % bowtie2 -x ./hg19/hg19 -1 ENCFF002DKA.fastq.gz -2 ENCFF002DKE.fastq.gz -S ENCFF002D
KA_DKE_Aligned.sam
stat: Bad file descriptor
Warning: Could not open read file "ENCFF002DKA.fastq.gz" for reading; skipping...
Error: No input read files were valid
(ERR): bowtie2-align exited with value 1
Mohana_Alla 🐼🐼🐼 ~ % bowtie2 -x ./hg19/hg19 -1 ENCFF002DKA.fastq -2 ENCFF002DKE.fastq -S ENCFF002DKA_DKE
_Aligned.sam
74096905 reads; of these:
  74096905 (100.00%) were paired; of these:
    18447072 (24.90%) aligned concordantly 0 times
    31206255 (42.12%) aligned concordantly exactly 1 time
    24443578 (32.99%) aligned concordantly >1 times
-----
    18447072 pairs aligned concordantly 0 times; of these:
      4038037 (21.89%) aligned discordantly 1 time
-----
    14409035 pairs aligned 0 times concordantly or discordantly; of these:
      28818070 mates make up the pairs; of these:
        18055514 (62.65%) aligned 0 times
        7068549 (24.53%) aligned exactly 1 time
        3694007 (12.82%) aligned >1 times
87.82% overall alignment rate
Mohana_Alla 🐼🐼🐼 ~ % █
```

```
Mohana_Alla 🐼🐼🐼 ~ % Bowtie2 -x ./hg19/hg19 -1 ENCFF000HQR.fastq.gz -2 ENCFF000HQP.fastq.gz -S ENCFF000HQR_HQP_Aligned.sam
128178110 reads; of these:
  128178110 (100.00%) were paired; of these:
    43301006 (33.78%) aligned concordantly 0 times
    54130430 (42.23%) aligned concordantly exactly 1 time
    30746674 (23.99%) aligned concordantly >1 times
-----
    43301006 pairs aligned concordantly 0 times; of these:
      12115047 (27.98%) aligned discordantly 1 time
-----
    31185959 pairs aligned 0 times concordantly or discordantly; of these:
      62371918 mates make up the pairs; of these:
        36672839 (58.80%) aligned 0 times
        18825930 (30.18%) aligned exactly 1 time
        6873149 (11.02%) aligned >1 times
85.69% overall alignment rate
Mohana_Alla 🐼🐼🐼 ~ % █
```

Used Samtools for K562 & MCF-7 filter uniquely mapped reads, sort, and index the BAM files

```
Mohana_Alla 🤖🤖🤖 ~ % samtools view -bS ENCFF002DKA_DKE_Aligned.sam > ENCFF002DKA_DKE_Aligned.bam
Mohana_Alla 🤖🤖🤖 ~ % samtools sort ENCFF002DKA_DKE_Aligned.bam -o ENCFF002DKA_DKE_Aligned_Sorted.bam
[bam_sort_core] merging from 27 files and 1 in-memory blocks...
Mohana_Alla 🤖🤖🤖 ~ % samtools index ENCFF002DKA_DKE_Aligned_Sorted.bam
```

```
Mohana_Alla 🤖🤖🤖 ~ % samtools view -bS ENCFF000HQR_HQP_Aligned.sam > ENCFF000HQR_HQP_Aligned.bam
Mohana_Alla 🤖🤖🤖 ~ % samtools view -q 30 -b ENCFF000HQR_HQP_Aligned.bam > ENCFF000HQR_HQP_UniquelyAligned.bam
Mohana_Alla 🤖🤖🤖 ~ % samtools sort ENCFF000HQR_HQP_UniquelyAligned.bam -o ENCFF000HQR_HQP_UniquelyAligned_Sorted.bam
[bam_sort_core] merging from 15 files and 1 in-memory blocks...
Mohana_Alla 🤖🤖🤖 ~ % samtools index ENCFF000HQR_HQP_UniquelyAligned_Sorted.bam
```

Downloading and Preparing the hg19 Annotation File from UCSC

```
zsh: command not found: -zsh
Mohana_Alla 🤖🤖🤖 ~ % wget https://hgdownload.cse.ucsc.edu/goldenpath/hg19/bigZips/genes/hg19.ncbiRefSeq.gtf.gz
--2025-03-06 14:20:31-- https://hgdownload.cse.ucsc.edu/goldenpath/hg19/bigZips/genes/hg19.ncbiRefSeq.gtf.gz
Resolving hgdownload.cse.ucsc.edu (hgdownload.cse.ucsc.edu)... 128.114.119.163
Connecting to hgdownload.cse.ucsc.edu (hgdownload.cse.ucsc.edu)[128.114.119.163]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 19718897 (19M) [application/x-gzip]
Saving to: 'hg19.ncbiRefSeq.gtf.gz'

hg19.ncbiRefSeq.gtf 100%[=====] 18.80M 7.97MB/s in 2.4s

2025-03-06 14:20:34 (7.97 MB/s) - 'hg19.ncbiRefSeq.gtf.gz' saved [19718897/19718897]

Mohana_Alla 🤖🤖🤖 ~ % gunzip hg19.ncbiRefSeq.gtf.gz
Mohana_Alla 🤖🤖🤖 ~ % mv hg19.ncbiRefSeq.gtf hg19_ncbiRefSeq_annotation.gtf
```

Running Cufflinks for Individual Gene Expression Analysis

Command

```
cufflinks -G hg19_ncbiRefSeq_annotation.gtf
ENCFF000HQR_HQP_UniquelyAligned_Sorted.bam -o
ENCFF000HQR_HQP_GeneExpression
```

- Assembles and quantifies gene expression for ENCFF000HQR_HQP_UniquelyAligned_Sorted.bam using hg19_ncbiRefSeq_annotation.gtf.
- Outputs results in the ENCFF000HQR_HQP_GeneExpression/ directory.

Command

```
cufflinks -G hg19_ncbiRefSeq_annotation.gtf
ENCFF002DKA_DKE_UniquelyAligned_Sorted.bam -o
ENCFF002DKA_DKE_GeneExpression
```

- Assembles and quantifies gene expression for ENCFF002DKA_DKE_UniquelyAligned_Sorted.bam using hg19_ncbiRefSeq_annotation.gtf.
- Outputs results in the ENCFF002DKA_DKE_GeneExpression/ directory.

```
zsh: command not found: -zsh
Mohana_Alla 🙄🙄🙄 ~ % cufflinks -G hg19_ncbiRefSeq_annotation.gtf ENCFF000HQR_HQP_UniquelyAligned_Sorted.bam -o ENCFF000HQR_HQP_GeneExpression
Warning: Could not connect to update server to verify current version. Please check at the Cufflinks website (http://cufflinks.cbc.umd.edu).
[03:42:29] Loading reference annotation.
[03:42:39] Inspecting reads and determining fragment length distribution.
> Processed 34880 loci. [*****] 100%
> Map Properties:
>   Normalized Map Mass: 68727972.00
>   Raw Map Mass: 68727972.00
>   Fragment Length Distribution: Empirical (learned)
>   Estimated Mean: 277.42
>   Estimated Std Dev: 34.28
[03:48:18] Estimating transcript abundances.
> Processed 34880 loci. [*****] 100%
```



```
(base) Mohana_Alla 🤔🤔🤔 % cufflinks -G hg19_ncbiRefSeq_annotation.gtf ENCF002DKA_DKE_UniquelyAligned_Sorted.bam -o ENCF002DKA_DKE_GeneExpression
Warning: Could not connect to update server to verify current version. Please check at the Cufflinks website (http://cufflinks.cbc.umd.edu).
[02:13:01] Loading reference annotation.
[02:13:10] Inspecting reads and determining fragment length distribution.
> Processed 34880 loci. [*****] 100%
> Map Properties:
>   Normalized Map Mass: 63506245.50
>   Raw Map Mass: 63506245.50
>   Fragment Length Distribution: Empirical (learned)
>     Estimated Mean: 175.23
>     Estimated Std Dev: 67.13
[02:41:08] Estimating transcript abundances.
> Processed 34880 loci. [*****] 100%
```

Using Cuffdiff to Calculate Differential Expression Between K562 and MCF7 Cells

```
cuffdiff -o ./K562_MCF7 hg19_ncbiRefSeq_annotation.gtf
ENCF002DKA_DKE_UniquelyAligned_Sorted.bam
ENCF000HQR_HQP_UniquelyAligned_Sorted.bam
```

- Compares gene expression between ENCF002DKA_DKE_UniquelyAligned_Sorted.bam (K562) and ENCF000HQR_HQP_UniquelyAligned_Sorted.bam (MCF7).
- Uses hg19_ncbiRefSeq_annotation.gtf for gene annotations.
- Saves output in the ./K562_MCF7/ directory.

```
Mohana_Alla 🤔🤔🤔 % cuffdiff -o ./K562_MCF7 hg19_ncbiRefSeq_annotation.gtf ENCF002DKA_DKE_UniquelyAligned_Sorted.bam ENCF000HQR_HQP_UniquelyAligned_Sorted.bam
Warning: Could not connect to update server to verify current version. Please check at the Cufflinks website (http://cufflinks.cbc.umd.edu).
[23:00:42] Loading reference annotation.
[23:00:51] Inspecting maps and determining fragment length distributions.
[23:38:42] Modeling fragment count overdispersion.
> Map Properties:
>   Normalized Map Mass: 62974574.08
>   Raw Map Mass: 30649384.00
>   Fragment Length Distribution: Empirical (learned)
>     Estimated Mean: 175.23
>     Estimated Std Dev: 67.13
> Map Properties:
>   Normalized Map Mass: 62974574.08
>   Raw Map Mass: 35813865.00
>   Fragment Length Distribution: Empirical (learned)
>     Estimated Mean: 277.42
>     Estimated Std Dev: 34.28
[23:38:43] Calculating preliminary abundance estimates
[23:38:43] Testing for differential expression and regulation in locus.
> Processed 34880 loci. [*****] 100%
Performed 37258 isoform-level transcription difference tests
Performed 0 tss-level transcription difference tests
Performed 21295 gene-level transcription difference tests
Performed 0 CDS-level transcription difference tests
Performed 0 splicing tests
Performed 0 promoter preference tests
Performing 0 relative CDS output tests
Writing isoform-level FPKM tracking
Writing TSS group-level FPKM tracking
Writing gene-level FPKM tracking
Writing CDS-level FPKM tracking
Writing isoform-level count tracking
Writing TSS group-level count tracking
Writing gene-level count tracking
Writing CDS-level count tracking
Writing isoform-level read group tracking
Writing TSS group-level read group tracking
Writing gene-level read group tracking
Writing CDS-level read group tracking
Writing read group info
Writing run info
Mohana_Alla 🤔🤔🤔 %
```

Procedures and Results

- **Alignment with Bowtie2:**

First, I downloaded the paired-end RNA-Seq reads for both K562 and MCF-7 cell lines from the ENCODE database.

I aligned these reads to the hg19 reference genome using Bowtie2, ensuring that each read pair was mapped to its correct genomic location.

This step is crucial for determining where each RNA fragment originated from within the genome.

- **Filtering with Samtools:**

After alignment, the output was in SAM format, which contained all mapped and unmapped reads.

I used Samtools to filter out only uniquely mapped reads, meaning each read had a single, best alignment to the genome.

This step helps remove ambiguous alignments and reduces false positives in the downstream expression analysis.

- **Sorting and Indexing the BAM Files:**

The filtered reads were then converted into BAM format and sorted using Samtools.

Sorting the BAM file makes it easier to process in downstream analyses, ensuring that all reads are arranged in order based on their genomic coordinates.

I also indexed the BAM files, which allows tools like Cufflinks and IGV (Integrative Genomics Viewer) to quickly access specific genomic regions for analysis.

- **Transcript Assembly and Gene Expression Estimation with Cufflinks:**

Next, I used Cufflinks to assemble the transcripts and estimate gene expression levels for each sample individually.

Cufflinks took the sorted BAM files and used a reference annotation file (hg19_ncbiRefSeq_annotation.gtf) to reconstruct known transcripts and quantify gene expression.

This step provided expression values in FPKM (Fragments Per Kilobase Million), giving an idea of how actively different genes are expressed in each cell line.

Differential Expression Analysis with Cuffdiff:

To compare the gene expression between K562 and MCF-7, I used Cuffdiff, which performs differential gene expression analysis.

Cuffdiff analyzed the gene expression differences between the two cell lines and calculated:

- Fold changes (how much a gene's expression increased or decreased).
- p-values and q-values (statistical measures to determine whether the changes are significant).

This step helped identify genes that are significantly upregulated or downregulated in one cell line compared to the other.

Results:

- **Gene Expression Estimates (from Cufflinks):**

Cufflinks provided a detailed report on gene expression levels for each sample.

The output included a `genes.fpkm_tracking` file, which contained expression values for each gene in FPKM (Fragments Per Kilobase Million).

This data shows which genes are highly expressed, moderately expressed, or lowly expressed in each cell line.

- **Differential Expression Results (from Cuffdiff):**

Cuffdiff produced a `gene_exp.diff` file, listing all genes with significant expression differences between K562 and MCF-7.

This file included:

- Fold changes (e.g., a gene might be expressed 5x higher in K562 than MCF-7).
- p-values and q-values to assess statistical significance.

These results help identify genes that may play key roles in the biological differences between the two cell lines.

Conclusion:

Through these steps, I was able to successfully align RNA-Seq reads, filter high-quality mapped reads, quantify gene expression, and identify differentially expressed genes between K562 and MCF-7. The final analysis provides biological insights into how these two cancer cell lines differ at the transcriptional level, which could help in understanding regulatory mechanisms or identifying potential therapeutic targets.